IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
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Executive Summary

- Methodology Overview
  - Data was gathered from two sources through API and web scraping.
  - Performed data cleaning and preparation.
  - Conducted exploratory data analysis using SQL and visual tools.
  - Developed interactive visualizations with dashboards and folium maps.
  - Applied machine learning classification models for predictive analysis. Models used include Logistic Regression, SVM, Decision Trees, and K Nearest Neighbors.
- Summary of Findings
  - SpaceX operates from four different launch sites.
  - Two booster versions, F9 v1.1 B1012 and F9 v1.1 B1015, failed to land on drone ships in 2015.
  - Interactive analytics revealed that launch sites are typically located in safe areas near the sea with strong logistical support.
  - Predictive analysis identified the Decision Tree Classifier as the most effective model for predicting successful landings, achieving over 89% accuracy and more than 90% accuracy on test data.

Introduction

- Project background and context

SpaceX's success comes from affordable rocket launches, with Falcon 9 costing $62 million compared to others at $165 million, mainly due to reusing the first stage. The Falcon 9 has three stages, with the large, costly first stage doing most work and sometimes landing successfully. However, it may crash or be sacrificed based on mission needs like payload and orbit. The goal of the project is to create a machine learning pipeline to predict the successful landing of the first stage.

- Problems you want to find answers
  - Which factors affect the landing success of the first stage?
  - How do these features interact?
  - What operating conditions are required to ensure a successful first stage landing?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Data was collected using SpaceX API and web scraping done on Wikipedia.
- Perform data wrangling
  - One hot encoding was done on categorical features.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Using a grid search for best parameters, classification models like logistic regression, SVM, decision tree and KNN classifiers were evaluated.

Data Collection

- Data collection was done using two methods

  1. Using the SpaceX API.

     - Use get request to collect the data.

     - The response content was received as a Json and was turned into a pandas dataframe.

  2. Web scraping of Wikipedia page.

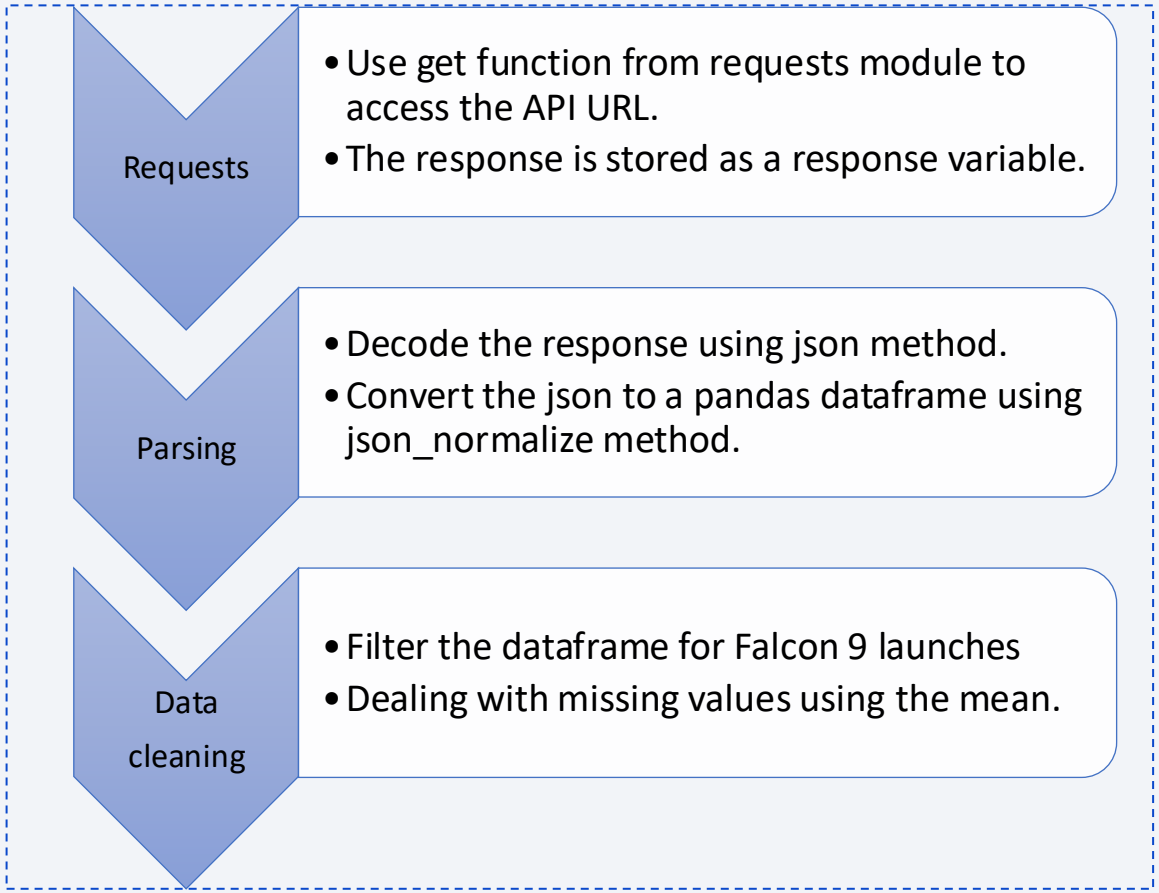
     - Wikipedia page for Falcon 9 launch data was scraped using BeautifulSoup.

     - The HTML table was extracted and parsed, which was then converted to a pandas dataframe.

Data Collection – SpaceX API

- Using Space X API to collect the launch data.

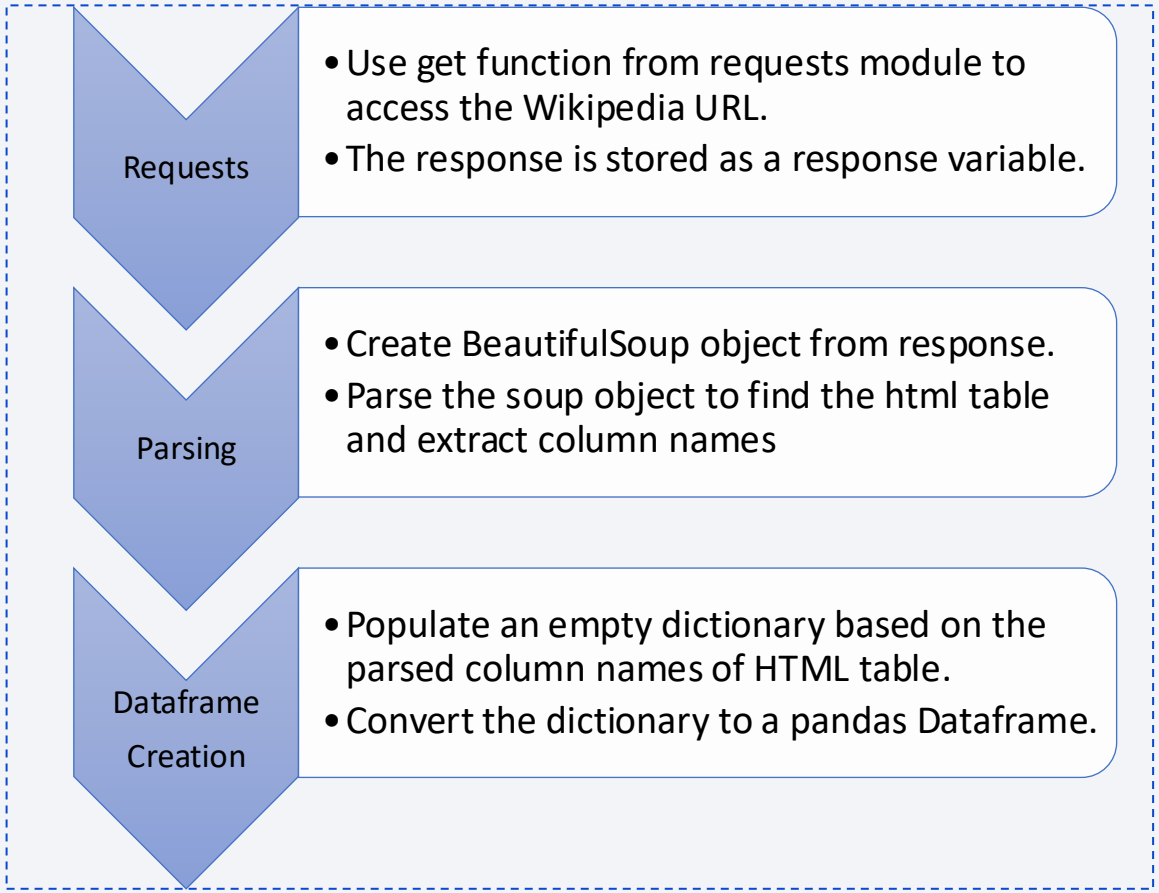
- This is the link to the notebook containing the data collection using SpaceX API.



Data Collection - Scraping

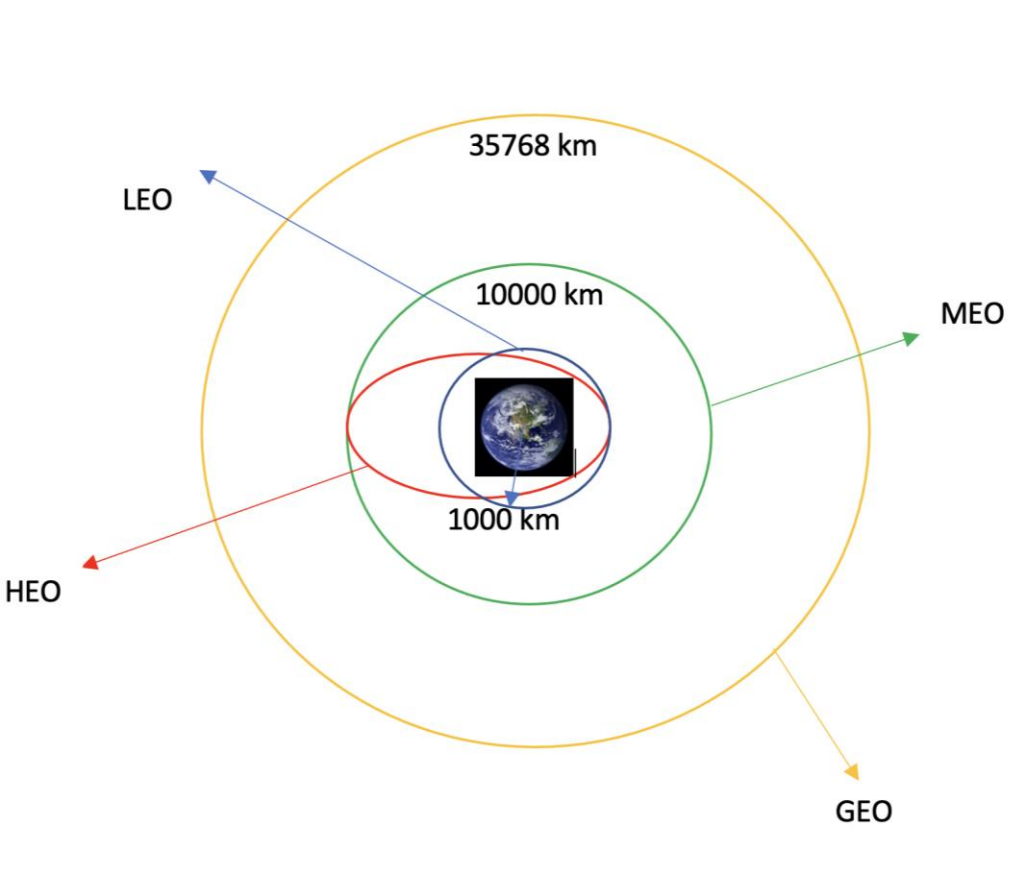
- Web scraping on Wikipedia page of Falcon 9 using BeautifulSoup.

- This is the link to the notebook containing the web scarping results of Falcon 9 Wikipedia page.



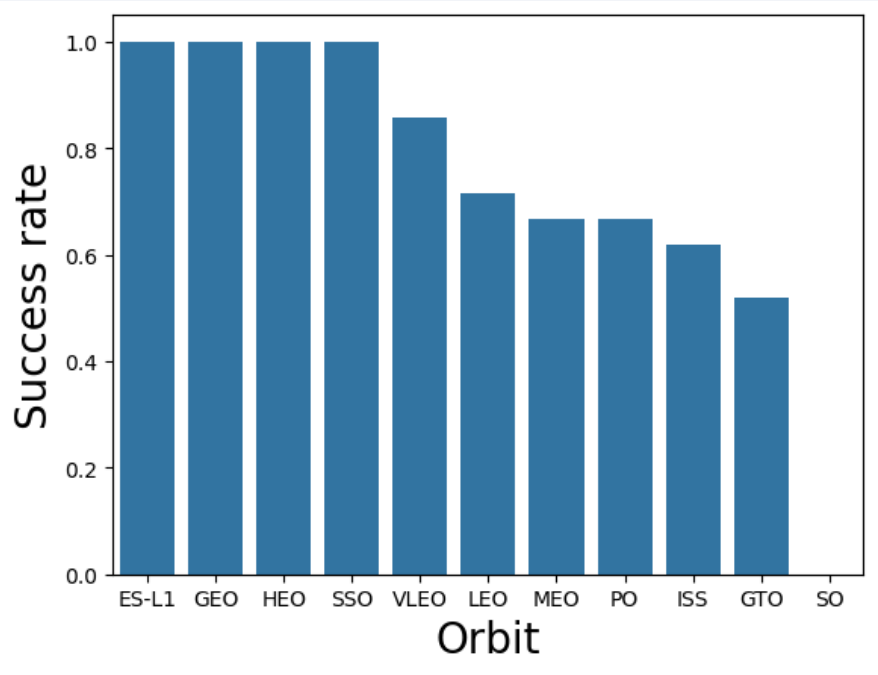
Data Wrangling

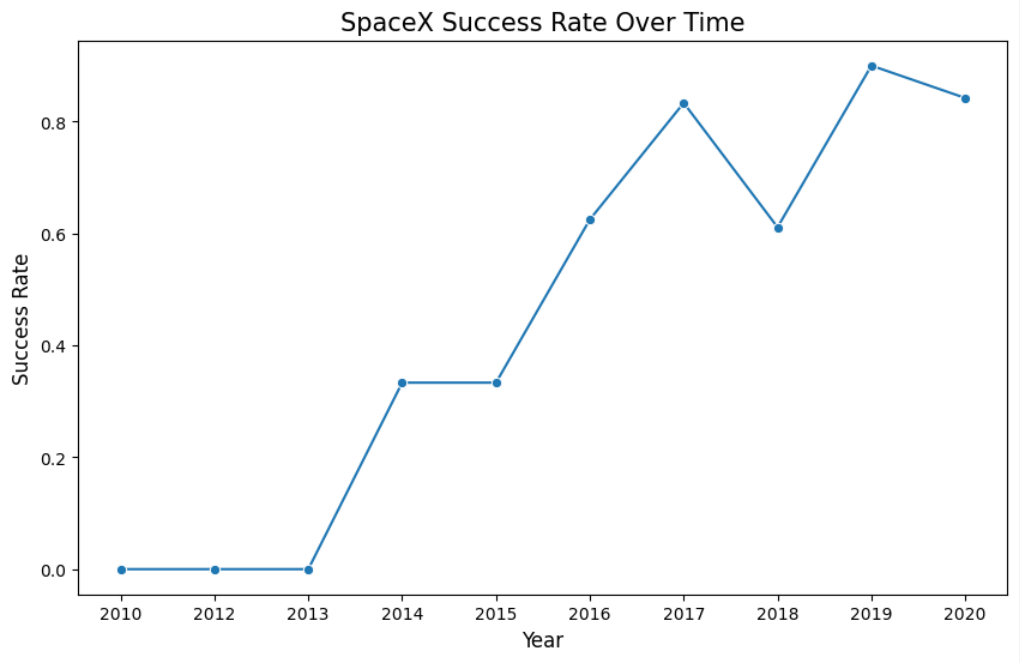
- Exploratory Data Analysis (EDA) to find some patterns in the data and determine what would be the label for training supervised models.
- Calculated the number of launches at each site and the number of launches to a particular orbit type.
- Created landing outcome labels from the mission outcomes.
- This is the link to the notebook detailing the EDA done on the dataset.



EDA with Data Visualization

- The relationships between various features such as Flight Number, Launch Site, Payload Mass, Orbit type were understood using scatter plots colored according to their success.
- Then success rates for different orbits were calculated and plotted as a bar chart.
- The time series plot using a line chart indicated that SpaceX success rate over the years increased.
- This is the link to the notebook for EDA with visualization.



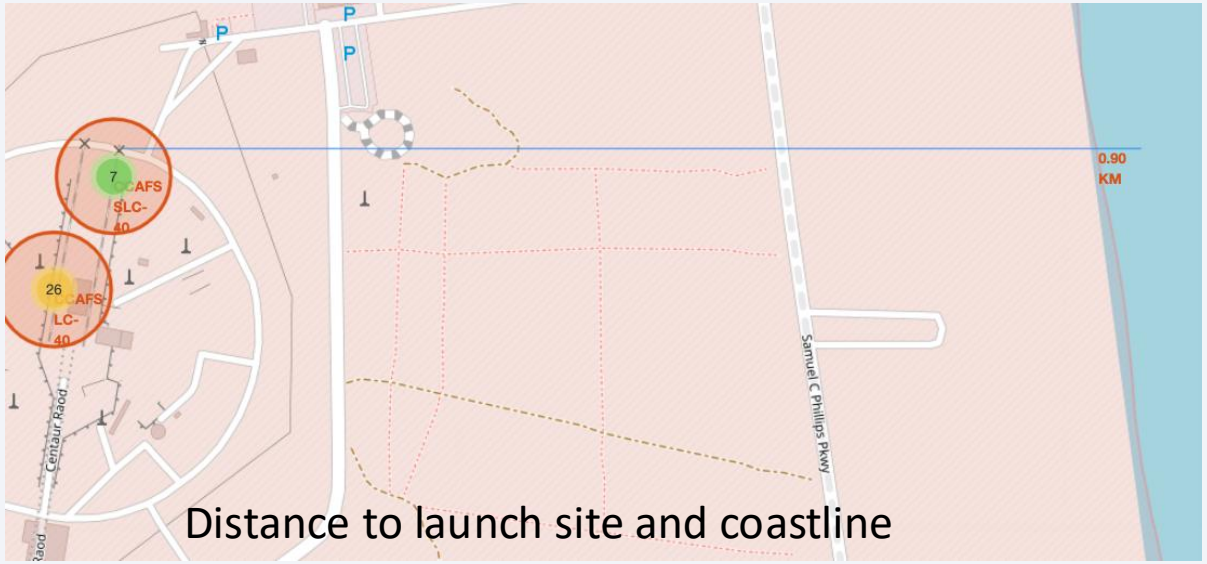


EDA with SQL

- EDA was performed with magic SQL commands in Jupyter notebook.
  - Display the names of the unique launch sites in the space mission
  - Display 5 records where launch sites begin with the string 'CCA'
  - Display the total payload mass carried by boosters launched by NASA (CRS)
  - Display average payload mass carried by booster version F9 v1.1
  - List the date when the first successful landing outcome in ground pad was achieved.
  - List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000.
  - List the total number of successful and failure mission outcomes
  - List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.
  - List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.
  - Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.
- This is the link to the EDA done using SQL magic commands in Jupyter notebook.

Build an Interactive Map with Folium

- Using map objects such as markers, circles, lines, etc. on a folium map all launch sites and their outcomes were added.
- Using marker clusters which are labelled with colors, the launch sites with high success rate were identified.
- This is the link to the notebook with folium map visualizations.



Distance to launch site and coastline

Build a Dashboard with Plotly Dash

- Using interactive dashboard, pie charts showing the total launches per site were added.
- A scatter plot showing the relationship of outcome and payload mass for different booster versions were added.
- Drop down menus were added to filter according to the launch site.
- This is the link to the python file for plotly dashboard.

Predictive Analysis (Classification)

- Loaded the data in to a pandas dataframe. After that standard scaling and splitting in to test and training sets were performed.
- Hyperparameters were optimized using gridsearch for linear regression, decision tree, SVM and KNN classifiers.
- The accuracy of the of the models were evaluated.
- This is the link to the notebook containing the predictive analysis done.

Data preprocessing

- Loaded data and standardized it.
- Split into training and testing sets

Hyperparameter Tuning

- Used gridsearch CV for hyperparameter tuning
- Best parameters found for linear regression, SVM, decision trees and KNN.

Evaluation

- Models evaluated with test datasets.
- Confusion matrices plotted for all the models.

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

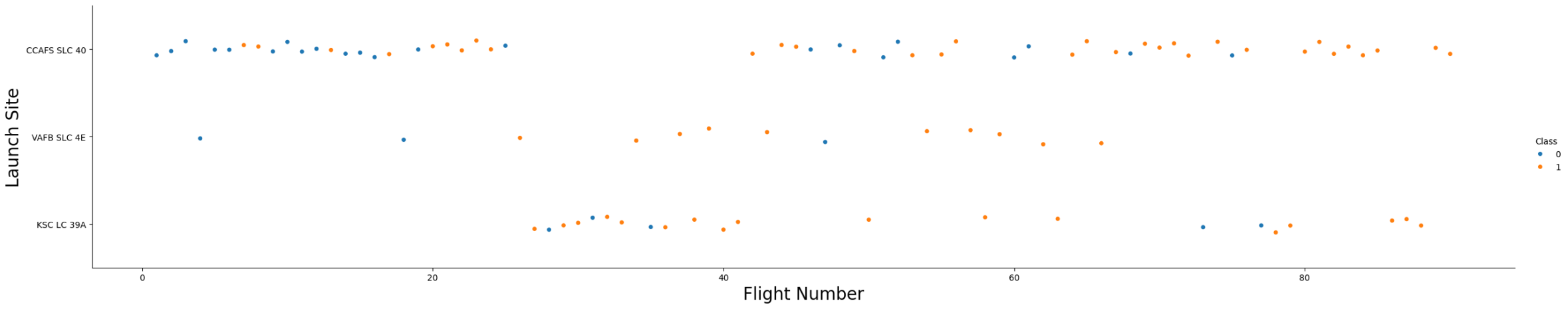
Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

- Scatter plot of Flight Number vs. Launch Site

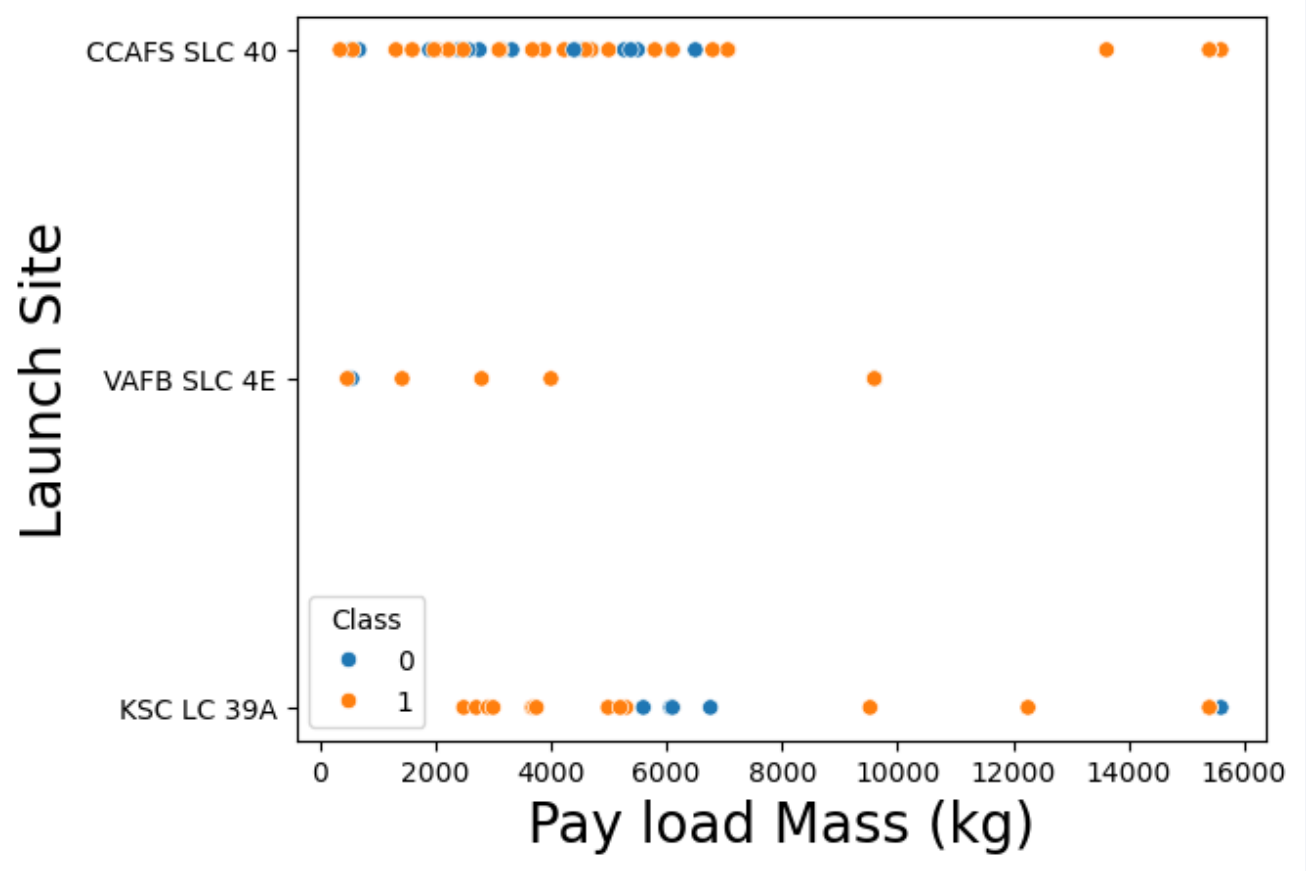
- Higher the flight number (continuous launch attempts) higher is the success rate.



Payload vs. Launch Site

- Scatter plot of Payload vs. Launch Site

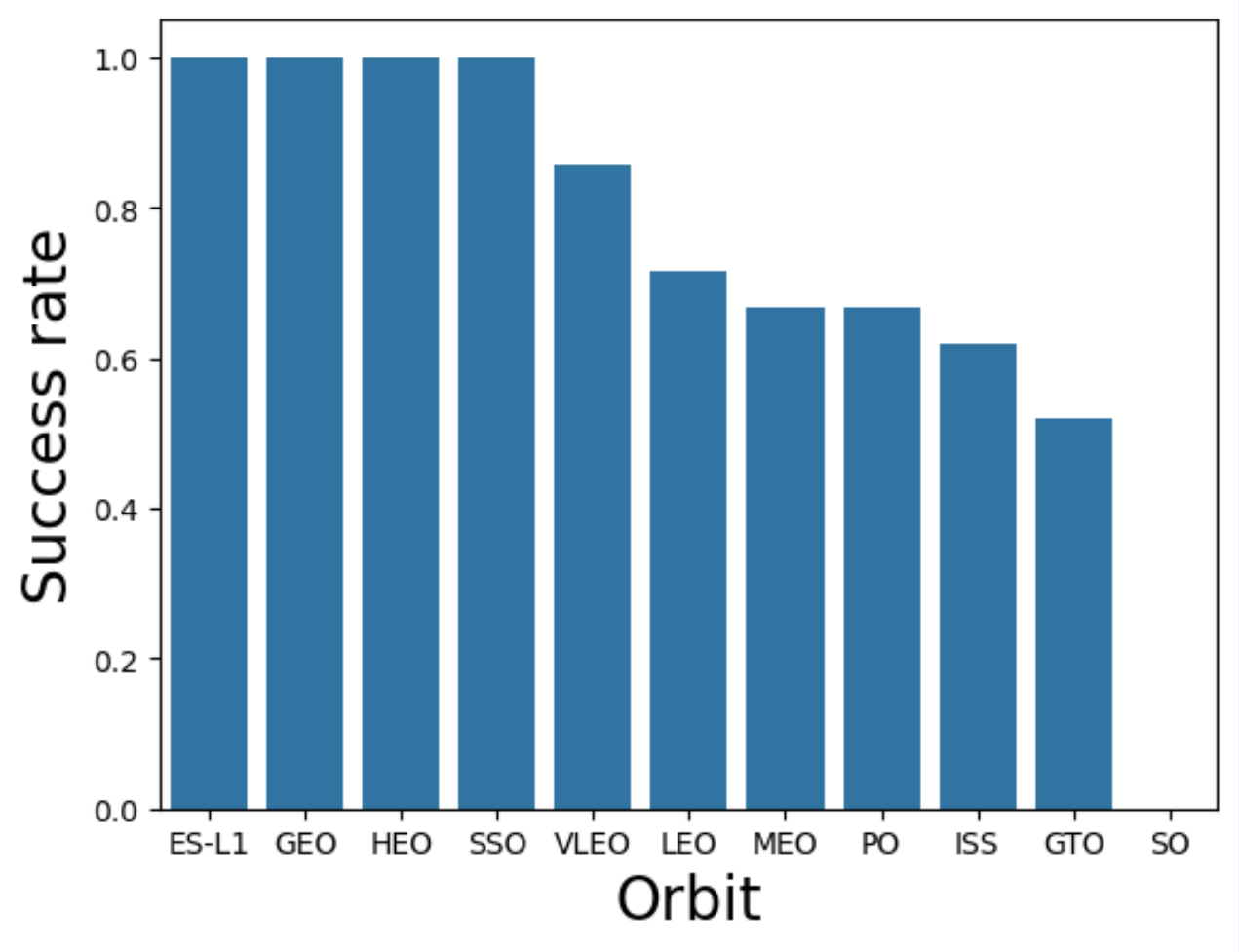
- We find for the VAFB-SLC launch site there are no rockets launched for heavy payload mass (greater than 10000).



Success Rate vs. Orbit Type

- Bar chart for the success rate of each orbit type

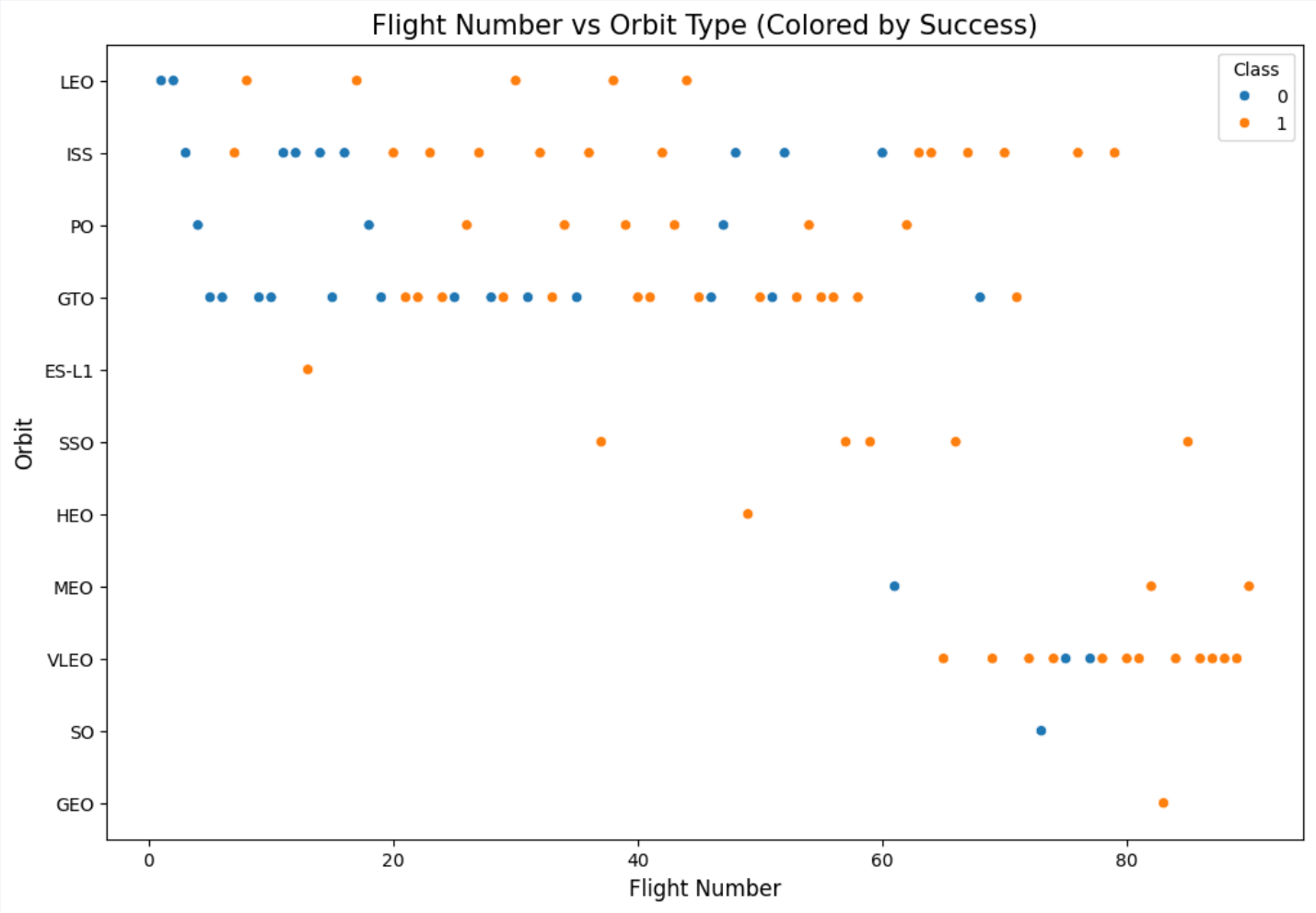
- ES-L1, GEO, HEO, SSO are the orbits with highest success rate followed by VLEO.



Flight Number vs. Orbit Type

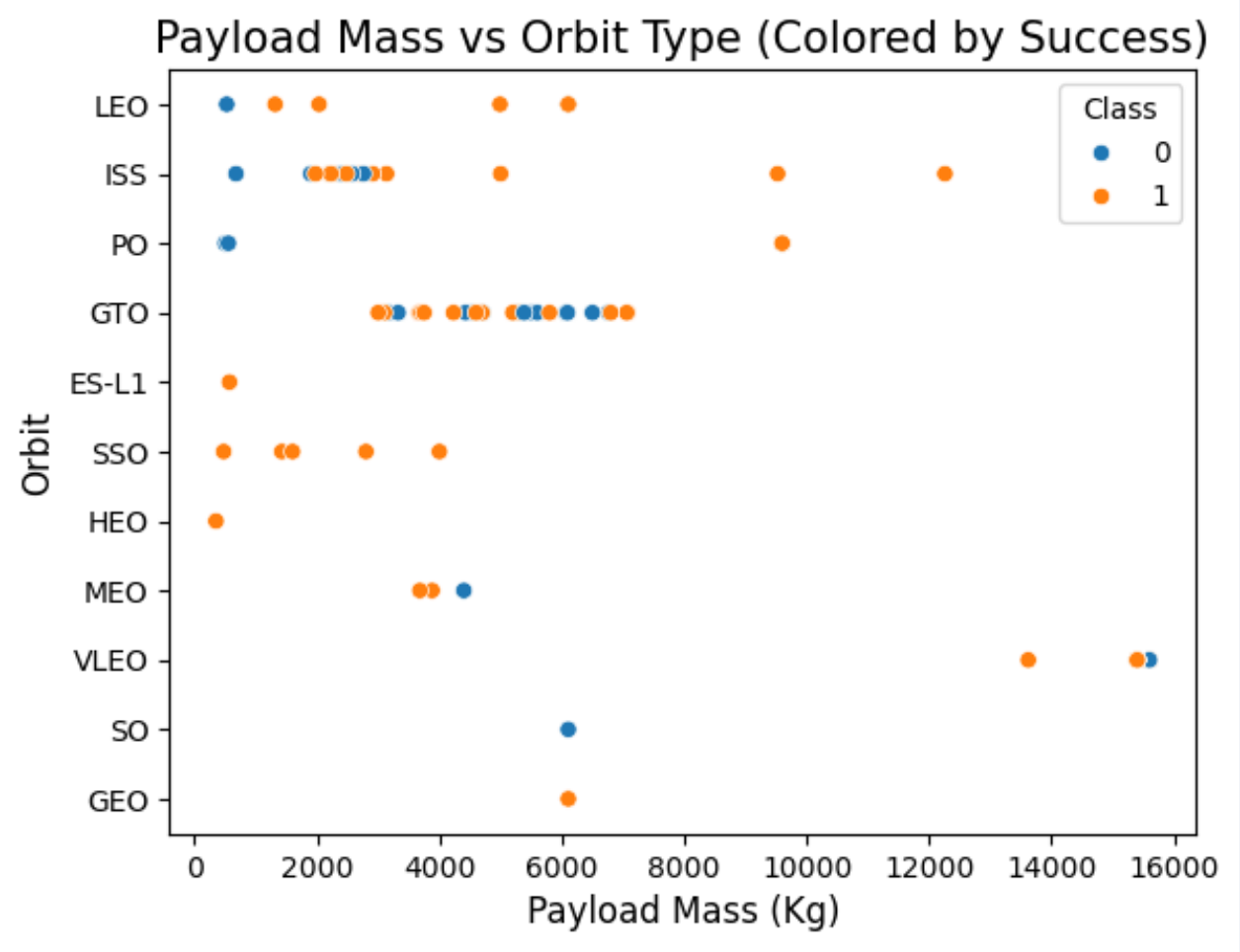
- Scatter point of Flight number vs. Orbit type

- In the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.



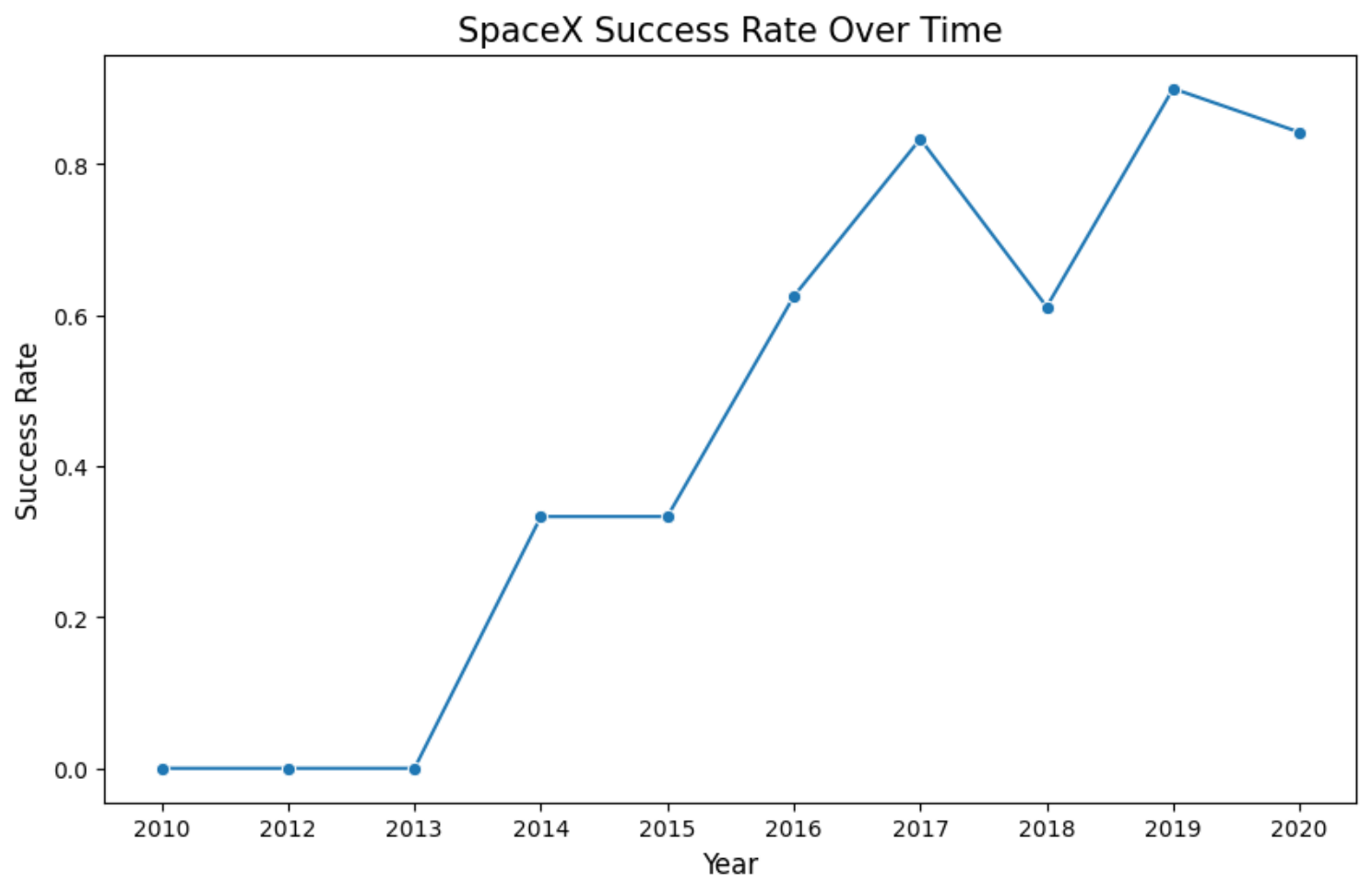
Payload vs. Orbit Type

- Scatter plot of payload vs. orbit type
- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.
- However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.



Launch Success Yearly Trend

- A line chart of yearly average success rate
- The success rate since 2013 kept increasing till 2020.



All Launch Site Names

- Use distinct to find the unique launch sites from the table.

```
%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with `CCA`
- %sql SELECT * FROM SPACEXTABLE WHERE Launch_Site Like "CCA%" LIMIT 5;

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Calculate the total payload carried by boosters from NASA
- %sql SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Customer Like "NASA%";
- 45596

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1
- %sql SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Booster_Version Like "F9 v1.1%";

AVG(PAYLOAD_MASS__KG_)
2534.6666666666665

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad
- %sql SELECT MIN(DATE) FROM SPACEXTABLE WHERE Landing_Outcome Like "Success (ground pad)";

MIN(DATE)
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

- %sql SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = "Success (drone ship)" AND PAYLOAD_MASS__KG_ < 6000 AND PAYLOAD_MASS__KG_ > 4000;

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes
- %sql SELECT Mission_Outcome, COUNT(*) AS Total_Count FROM SPACEXTABLE GROUP BY Mission_Outcome;

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass
- %sql SELECT Booster_Version, PAYLOAD_MASS__KG_ FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
- In January and in April there were failures with drone ship in yrat 2015

Month_Name	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

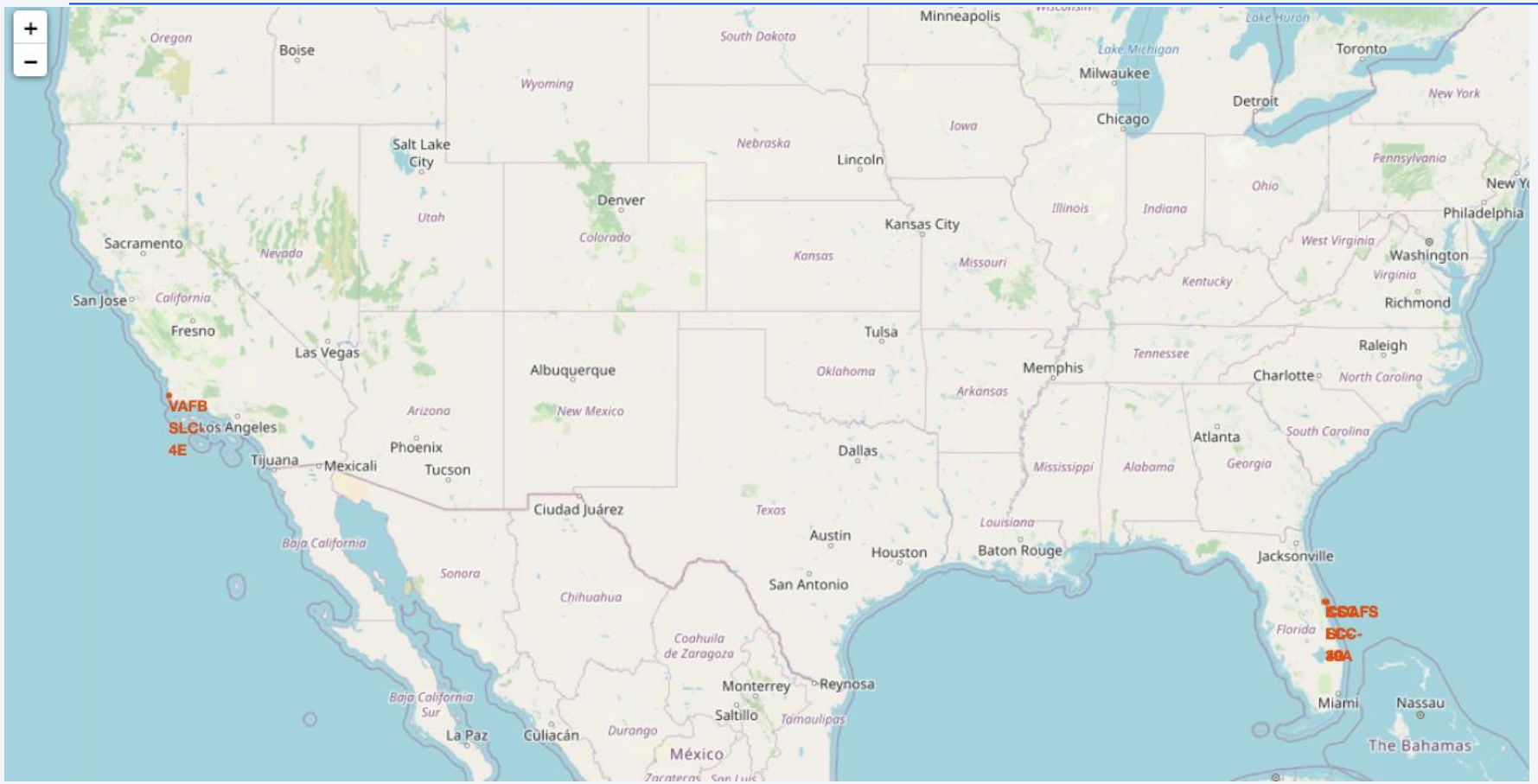
- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
- %sql SELECT Landing_Outcome, COUNT(*) AS Outcome_Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Outcome_Count DESC;

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

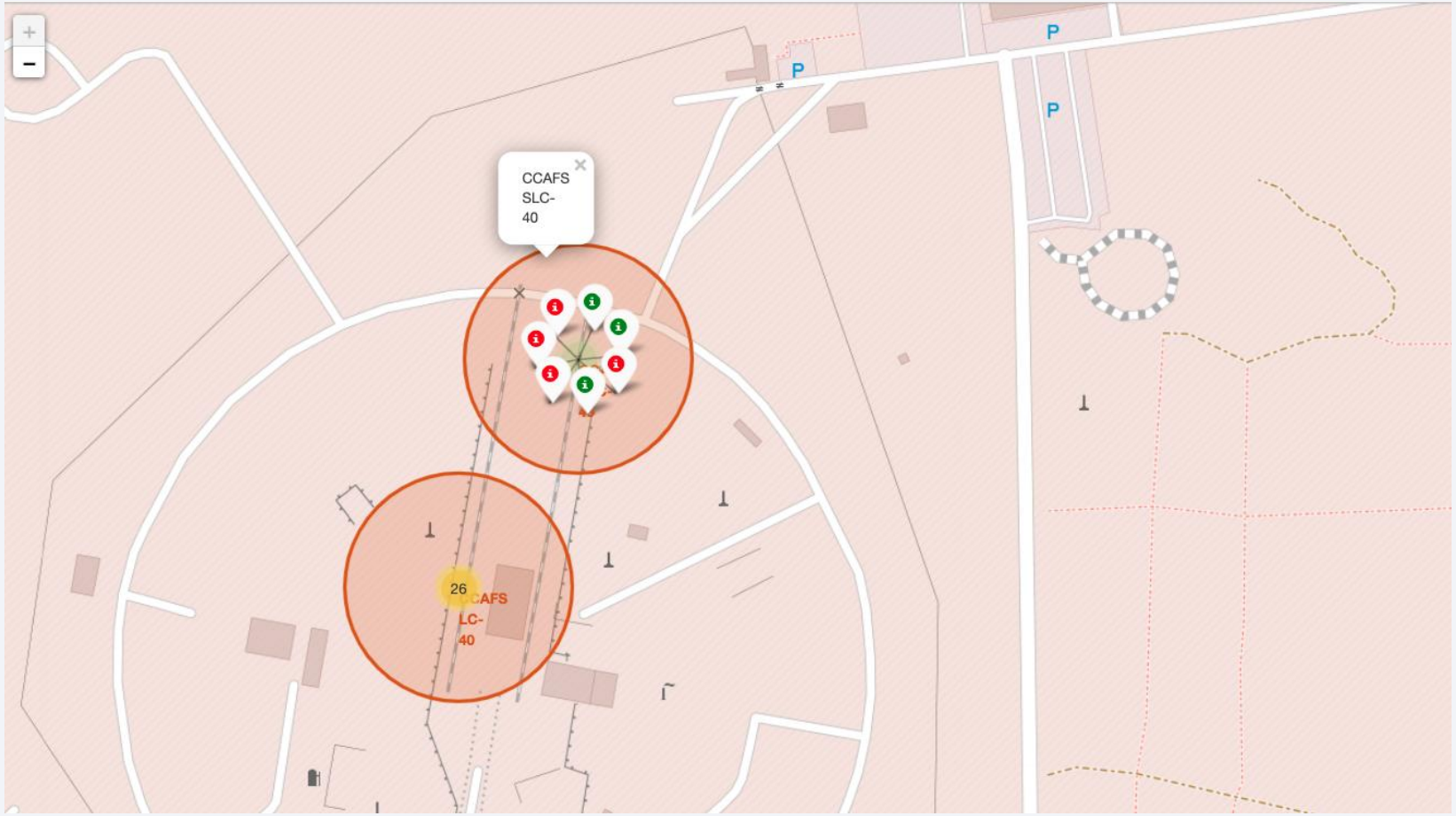
Launch Sites Proximities Analysis

All Launch Site



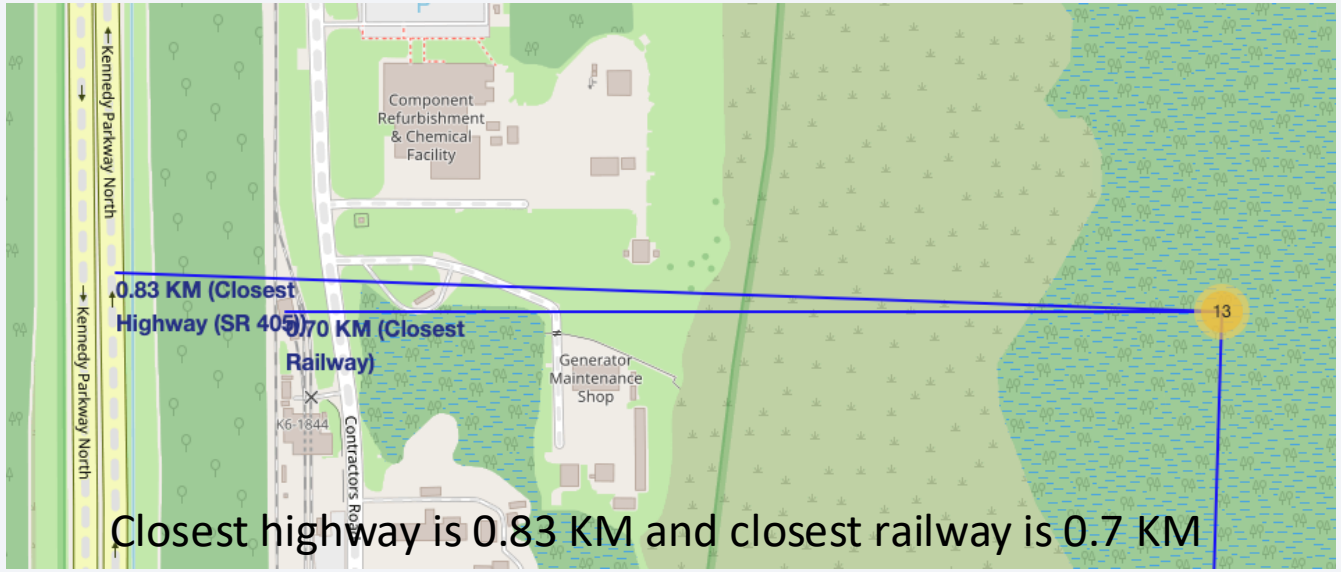
Space X launch sites are located in united states close to the coastline.

Markers showing Launch sites with their outcome

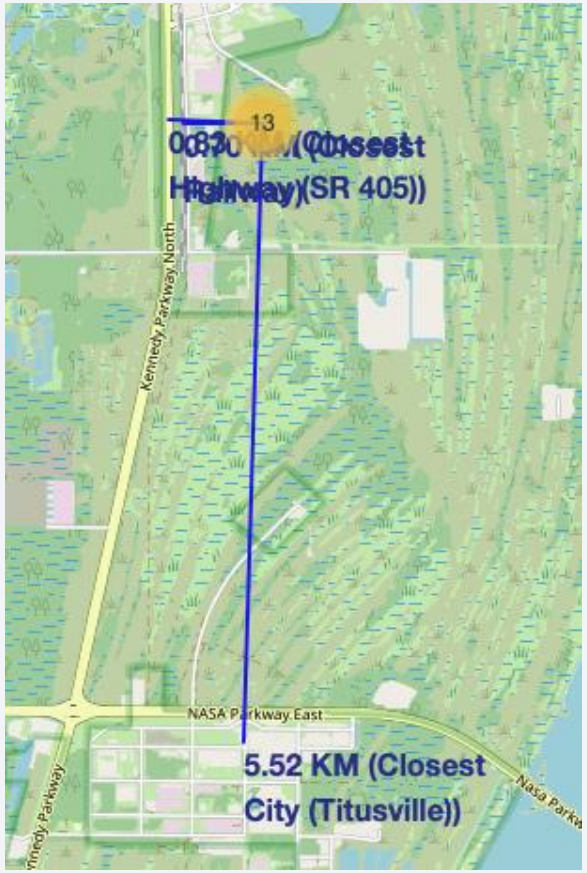


Launch sites where the green marker shows a successful launch and red shows a failure.

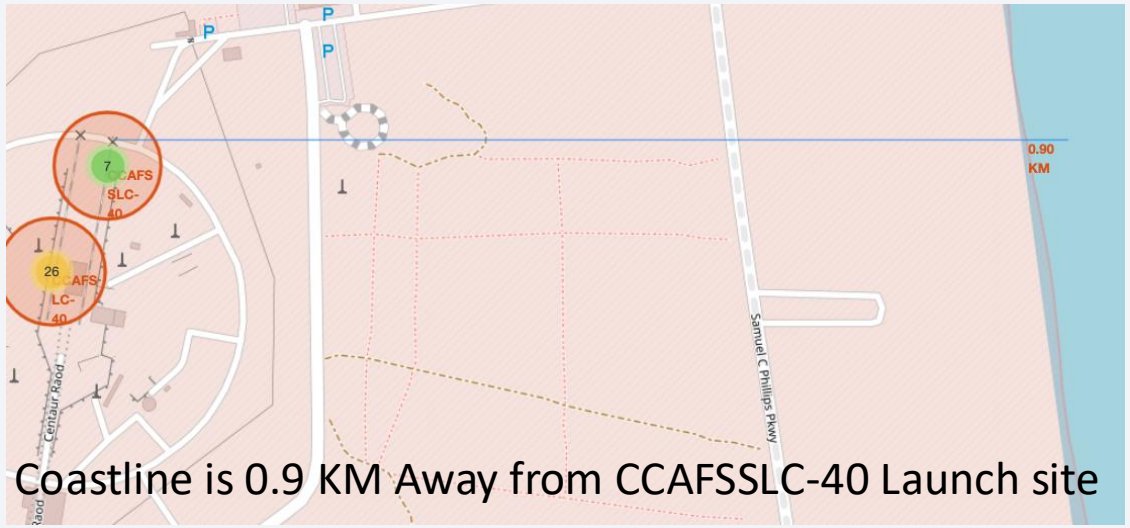
Launch site proximity with landmarks



Closest highway is 0.83 KM and closest railway is 0.7 KM



Closest city Titusville is 5.52km away



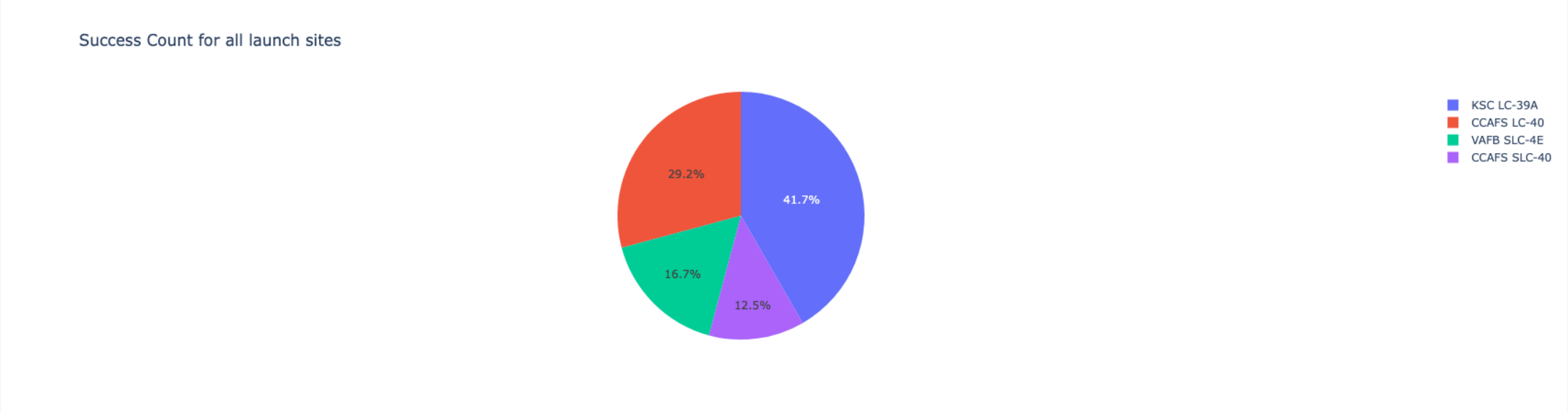
Coastline is 0.9 KM Away from CCAFSSLC-40 Launch site

Section 4

Build a Dashboard with Plotly Dash

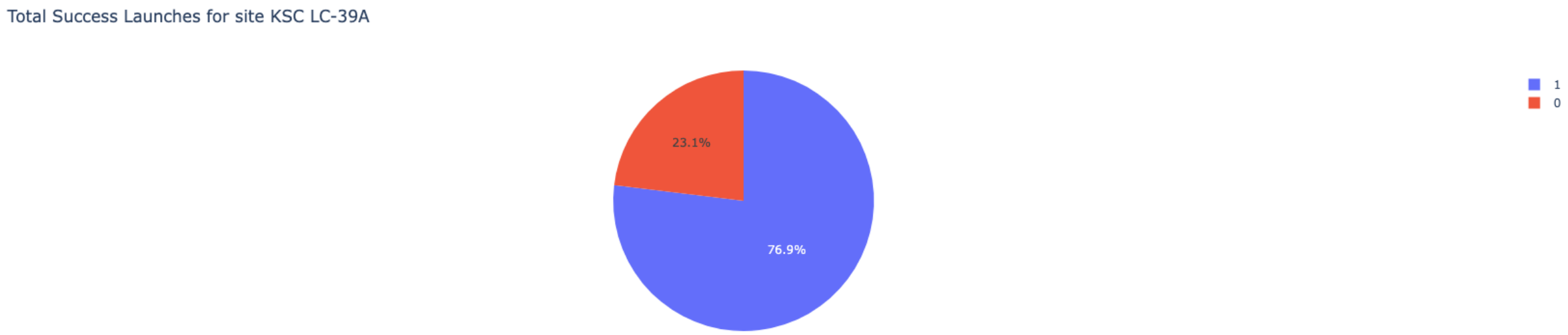
Success count for all launch sites

- KSC LC-39A had the greatest number of successes.



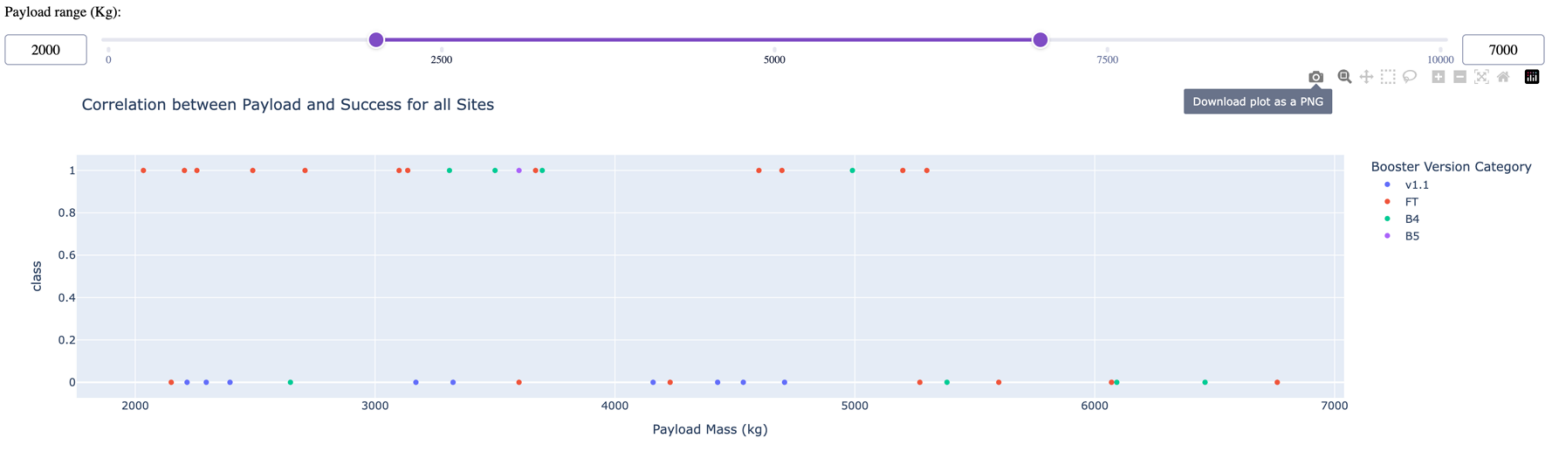
Launch success rate for KSC LC-39A

KSC LC-39A had a success rate of 76.9%



Correlation between Payload mass and success

- Success for lower payload mass is higher than for heavier payloads.



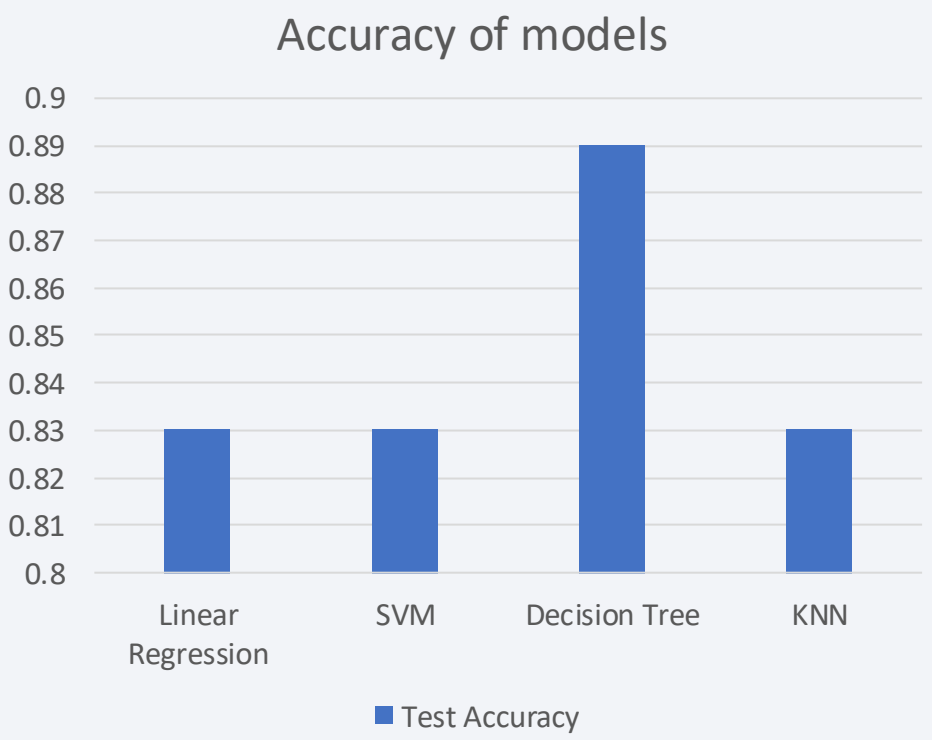
Section 5

Predictive Analysis (Classification)

Classification Accuracy

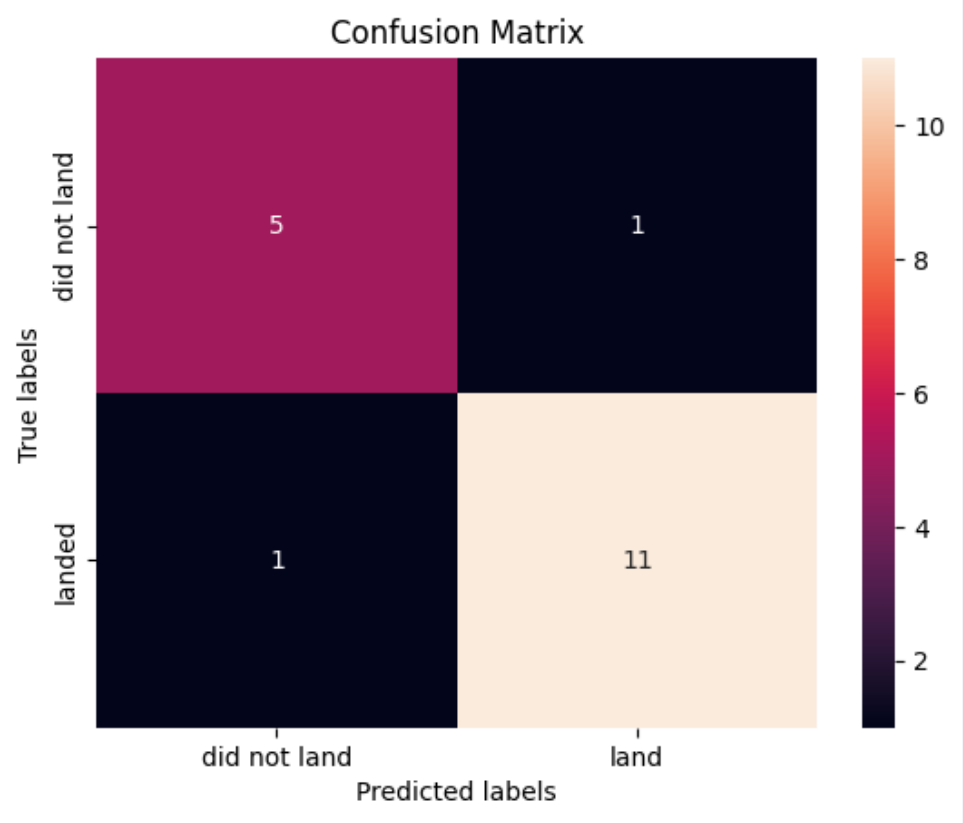
- The accuracy of the all models are plotted in a bar graph.

- The decision tree is the best classifier.



Confusion Matrix

- Confusion matrix of the decision tree with highest accuracy. The decision tree only has one false positive.



Conclusions

- The larger the number of flights at a launch site , the greater is the success rate.
- Launch success rate increased steadily as the years passed.
- Orbits ES-L1, GEO, HEO, SS, VLEO had the highest success rates,
- KSC LC-39A had the highest number of successful launches.
- A decision tree classifier would be the best model to predict if a given launch would be successful.

Appendix

- The github URL for the project is https://github.com/skatnagallu/SpaceY_DataScience_Capstone_Project.git

Thank you!